MCA 1st Semester Examination, 2023 (R & B)

Subject : MCA

(*Data Structure*)

Paper —1.1.4

Full Marks : 80

Time : 3 hours

Answer **all** questions

The figures in the right-hand margin indicate marks

Candidates are required to give their answers in their own words as far as practicable

1. (*a*) Explain ADT with it's type and features. Write down necessity of space and time complexity with example. 10

   (*b*) What is single linked list ? Write algorithm to insert element at end of single linked list. 10

(*Turn Over*)

Or

(*c*) What is an arrary ? Write down types of array with example. Write down the row and column major order of following matrices. 10

int $a[2][3] = \{\{1, 2, 3\}, \{4, 5, 6)\}\}$;

(*d*) What is queue ? Explain operations performed on queue. Write down algorithm for insert, delete and display operation in queue. 10

2. (*a*) What is binary tree traversal ? Explain different types of binary tree traversal with example ? Write down algorithm for inorder, pre-order and post-order traversal. 10

(*b*) Write down various types of tree. Explain in detail with example and algorithm of binary search tree. 10

Or

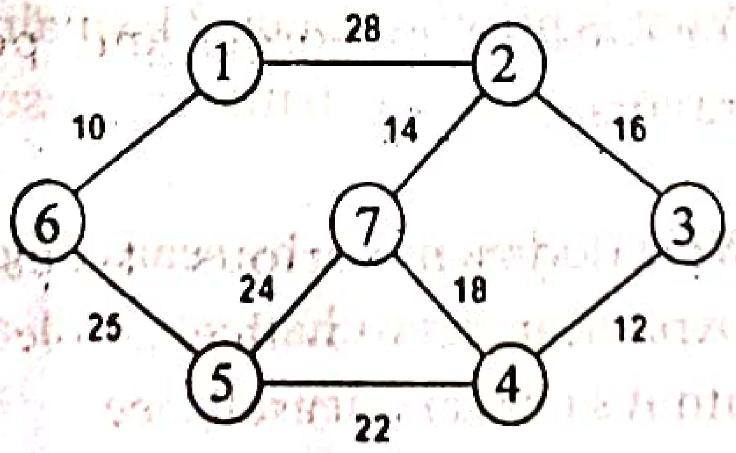
(*c*) Write down use and implementation of expression tree. Create expression tree for 3 + ((5 + 9)*2). 10

(*d*) Explain Red Black tree in detail. Create a RED BLACK Tree by inserting following sequence of number 10

8, 18, 5, 15, 17, 25, 40 & 80.

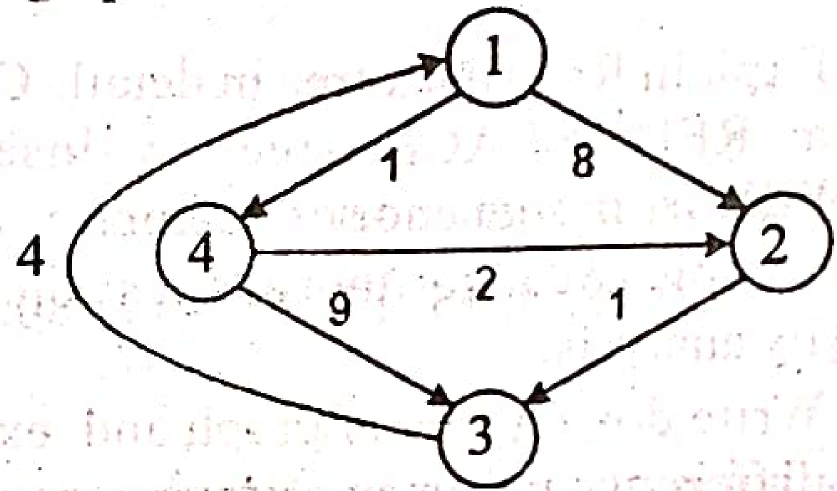
3. (*a*) Write down types of graph and explain different types of graph traversal mechanism with example. 10

(*b*) Construct the minimum spanning tree (MST) for the given graph using Prim's Algorithm. 10



Or

(*c*) Consider the following directed weighted graph



Using Floyed Warshall Algorithm, find the shortest path distance between every pair of vertices. 10

(*d*) What is priority queue ? Explain types of priority queue with their representation. 10

4. (*a*) What do you mean by searching ? Explain linear search mechanism with algorithm and it's complexity analysis. 10

(*b*) Explain with example about need of sorting. Write algorithm with example about bubble sort and it's complexity analysis. 10

Or

(*c*) What is merge sort ? Explain with example and algorithm and it's complexity analysis. 10

(*d*) What is dynamic memory management ? Explain with example. Differentiate between garbage collection and compaction. 10